

Peer Response

by [Abdulla Husain Salem Hadna Almessabi](#) - Monday, 1 September 2025, 4:48 AM

Hi Ali,

Your post gives a concise description of the trade-offs between ACLs such as KQML and the traditional approach to method invocation. One of the greatest strengths of the ACLs is that they allow the standardisation of communication between a heterogeneous and distributed system (Moore, 2000). This is crucial in such areas as e-commerce and supply-chain management, where various agents, usually designed with different architectures, still need to work together (Labrou and Finin, 1998). The scalability feature you discussed matters, too, as multi-agent systems can typically be expected to dynamically scale by introducing new agents without re-architecturing the entire communication layer.

Your criticism of ACL limitations was also quite good in my opinion. Their parsing and reasoning overhead can often render them impractical for time-sensitive applications like robotics. Finin et al. (1994) stated that, though conceptually powerful, the cost of ACLs is less compelling in real-time scenarios. This tension is well brought out in your comparison of Python and Java's approach to method invocation: direct calls are fast but do not provide the open system freedom and abstraction.

Another possible approach might be the incorporation of the ACL principles in contemporary middleware. As an example, standards like the JADE or FIPA standards are trying to fill this gap between ACL-style semantics and more lightweight message-passing infrastructures (Bouzoubaa and Moulin, 1998). Such a hybrid solution can contribute to reducing the efficiency issues without loss of standardised agent interaction advantages (Prymek and Horak, 2008).

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by [Abdulla Husain Salem Hadna Almessabi](#) - Monday, 1 September 2025, 4:46 AM

Hi Fahad,

I concur with your focus on the semantics richness of ACLs like KQML, especially that they have a speech act theory underpinning. This design renders them much more than mere containers of data, as they transfer intention and societal commitments among agents (Chalupsky et al., 1992). ACLs are indeed well adapted to open, distributed environments where agents in different programming languages are to communicate with each other. Such interoperability is not something that method invocation in either Python or Java can naturally offer, because these two approaches are explicitly tied to a particular runtime (Finin et al., 1994). I also like the fact that you point out the computational cost and how a common ontology is needed. In the absence of shared language, agents can communicate with each other and yet be unable to reach a level of true understanding, defeating the benefits of semantic communication (Liu, Zhang, and Wu, 2011). That is why, as you pointed out, ACLs are still more common in the realm of academia and the lab, but not in the realm of commerce, where speed and reliability are more important.

Maybe one more aspect that should be taken into consideration is the increasing role of hybrid approaches. Many modern multi-agent systems combine lightweight APIs for efficiency with ACL-inspired protocols for higher-level reasoning. As an example, ontologies may be combined with conventional communication channels through such frameworks as JADE, which simplifies the process of uniting performance and semantics (Luo, Zou, and Luo, 2012). This indicates that, opposed to considering ACLs and method invocation as exclusive mechanisms, there is a possibility that the future of adaptive systems is to use both based on the circumstances.

References

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